SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 11

AIM : Reshaping data using pivot_longer()/pivot_wider() (R).

Input :

```
# R Script: Reshaping Data with pivot_longer() and pivot_wider()
# Dataset: loan_approval_dataset.csv
library(dplyr)
library(tidyr)
# 1. SETUP: Import Data
df <- read.csv("loan_approval_dataset.csv", na.strings = c("", "NA"))

# Add a unique ID for tracking during reshaping
df <- df %>% mutate(Applicant_ID = row_number())
cat("\n--- 1. Original Wide Data (First 6 Rows) ---\n")
print(head(df))

# 2. PIVOT_LONGER (Wide to Long)
# Automatically detect numeric columns for reshaping
num_cols <- names(df)[sapply(df, is.numeric)]
# Remove the ID column from pivoting
num_cols <- setdiff(num_cols, "Applicant_ID")
long_df <- df %>%
  pivot_longer(
    cols = all_of(num_cols),
    names_to = "Metric",
    values_to = "Value"
  )
cat("\n--- 2. Long Format (pivot_longer) ---\n")
print(head(long_df, 10))

# 3. PIVOT_WIDER (Long to Wide)
wide_df <- long_df %>%
```

```
  pivot_wider(
    names_from = Metric,
    values_from = Value
  )
cat("\n--- 3. Wide Format (pivot_wider - Back to Original) ---\n")
print(head(wide_df))

# 4. ADVANCED EXAMPLE – Category-wise Loan Report
# Choose a category column safely (Gender / Education / Property_Area)
cat_cols <- names(df)[sapply(df, is.character)]
# Pick the first category column automatically
category_col <- cat_cols[1]
cat("\n--- Category used for reporting ---\n")
print(category_col)
category_pivot <- df %>%
  select(Applicant_ID, all_of(category_col), all_of(num_cols[1])) %>%
  pivot_wider(
    names_from = all_of(category_col),
    values_from = all_of(num_cols[1])
  )
cat("\n--- 4. Category-based Pivot Table ---\n")
print(head(category_pivot))
```

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Output :